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logistic_regression.py

```
import numpy as np
from sklearn.linear_model import LogisticRegression
# Study hours
X = np.array([[1], [2], [3], [4], [5], [6], [7], [8]])
# Result: 0 = Fail, 1 = Pass
y = np.array([0, 0, 0, 0, 1, 1, 1, 1])
# Create model
model = LogisticRegression()
# Train model
model.fit(X, y)
# Predict for a student who studied for 6 hours
prediction = model.predict([[6]])
print("Prediction:", prediction)
# Probability
probability = model.predict_proba([[6]])
print("Probability:", probability)
```

Output:

```
(venv) PS C:\Users\Jites\Desktop\New folder> & "c:\Users\Jites\Desktop\New folder\venv\Scripts\python.exe" "c:/Users/Jites/Desktop/New folder/logistic_regression.py"
Prediction: [1]
Probability: [[0.14745684 0.85254316]]
(venv) PS C:\Users\Jites\Desktop\New folder> █
```

logistic_regression_train_test.py

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
# Dataset
X = np.array([[1], [2], [3], [4], [5], [6], [7], [8],[9],[10]])
y = np.array([0, 0, 0, 0, 0, 1, 1, 1, 1, 1])
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create model
model = LogisticRegression()
# Train model
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Predictions:", y_pred)
print("Actual:", y_test)
print("Accuracy:", accuracy)
# New student
new_student = [[7]]
```

```
prediction = model.predict(new_student)
probability = model.predict_proba(new_student)
print("Prediction:", prediction)
print("Probability:", probability)
```

Output:

```
(venv) PS C:\Users\Jites\Desktop\New folder> & "c:\Users\Jites\Desktop\New folder\venv\Scripts\python.exe" "c:/Users/Jites/Desktop/New folder/logistic_regression_train_test.py"
Predictions: [1 0]
Actual: [1 0]
Accuracy: 1.0
Prediction: [1]
Probability: [[0.15264473 0.84735527]]
```